



AVOGRA 100MG TABLETS

COMPOSITION:

Each tablet contains:
Sildenafil citrate 140.5 mg
equivalent to 100mg of Sildenafil

PRODUCT DESCRIPTION:

Avogra 100mg Tablet is a white, oval-shaped film coated tablet, engraved "NK" in one side and double "507" on the other side.

PHARMACOLOGICAL PROPERTIES:

Pharmacodynamic properties

ATC code : G04B E03

Sildenafil is a selective inhibitor of phosphodiesterases (PDEs), with the greatest selectivity for PDE type 5. At least 10 isoenzyme families of PDE have been identified, several of which (e.g. PDE types 5 and 6) selectively hydrolyze cyclic guanosine monophosphate (cGMP) relative to cyclic adenosine monophosphate (cAMP). Sildenafil selectively inhibits cGMP-specific PDE type 5, the principal isoenzyme involved in the metabolism of cGMP to GMP in the corpora cavernosa of the penis and clitoris, PDE types 2 and 3 also are present in these corpora cavernosa, and sildenafil also inhibits these isoenzymes albeit substantially less potently (i.e., requires doses exceeding usual therapeutic levels). PDE type 5 also has been isolated from lung, platelets, kidney, spleen and various vascular (e.g., penile and clitoral corpora cavernosa) and visceral smooth muscle (e.g., gastric fundus, esophageal sphincter, colon) and skeletal muscle but not from cardiac muscle. By inhibiting PDE type 5, sildenafil causes accumulation of cGMP in various tissues, including the penile and clitoral corpora cavernosa. The role of cGMP as a modulator of cAMP signal transduction pathways remains to be more fully elucidated, but *ex vivo* studies indicate that sildenafil does not appear to appreciably affect cAMP tissue concentrations.

Sildenafil also exhibits some activity against other PDE isoenzymes. In vitro, sildenafil is about 10 times more active against PDE type 5 than against PDE type 6, greater than 70-80 times more active against PDE type 5 than against PDE type 1, greater than 1000 times more active against PDE type 5 than against PDE types 2 and 4, and 4000 times more active against PDE type 5 than against PDE type 3. PDE type 3 is involved in cardiac contractility, vascular smooth muscle relaxation (vasodilation), and platelet aggregation, and PDE type 6 (photoreceptor PDE) is found in the retina and involved in phototransduction.

PHARMACOKINETIC PROPERTIES

Absorption

Sildenafil is rapidly absorbed. Maximum observed plasma concentrations are reached within 30 to 120 minutes (median 60 minutes) of oral dosing in the fasted state. The mean absolute oral bioavailability is 41 % (range 25-63 %). After oral dosing of sildenafil AUC and C_{max} increase in proportion with dose over the recommended dose range (25-100 mg).

When sildenafil is taken with food, the rate of absorption is reduced with a mean delay in t_{max} of 60 minutes and a mean reduction in C_{max} of 29%.

Distribution

The mean steady state volume of distribution (V_d) for sildenafil is 105 l, indicating distribution into the tissues. After a single oral dose of 100 mg, the mean maximum total plasma concentration of sildenafil is approximately 440 ng/ml (CV 40 %). Since sildenafil (and its major circulating N-desmethyl metabolite) is 96 % bound to plasma proteins, this results in the mean maximum free plasma concentration for sildenafil of

18 ng/ml (38 nM). Protein binding is independent of total drug concentrations.

Metabolism

Sildenafil is cleared predominantly by the CYP3A4 (major route) and CYP2C9 (minor route) hepatic microsomal isoenzymes. The major circulating metabolite results from N-demethylation of sildenafil. This metabolite has a phosphodiesterase selectivity profile similar to sildenafil and an *in vitro* potency for PDE5 approximately 50 % that of the parent drug. Plasma concentrations of this metabolite are approximately 40 % of those seen for sildenafil. The N-desmethyl metabolite is further metabolised, with a terminal half-life of approximately 4 h.

Elimination

The total body clearance of sildenafil is 41 l/h with a resultant terminal phase half-life of 3-5 h. After either oral or intravenous administration, sildenafil is excreted as metabolites predominantly in the faeces (approximately 80 % of administered oral dose) and to a lesser extent in the urine (approximately 13 % of administered oral dose).

INDICATIONS:

Treatment of erectile dysfunction, which is the inability to achieve or maintain a penile erection sufficient for satisfactory sexual performance. In order for sildenafil to be effective, sexual stimulation is required.

RECOMMENDED DOSE

For oral administration.

Adults: The recommended dose is 50 mg taken, as needed, approximately 1 hr before sexual activity. Based on effectiveness and tolerance, the dose may be increased to a maximum recommended dose of 100 mg or decreased to 25 mg. The maximum recommended daily dose is 100 mg. The maximum recommended dosing frequency is once per day.

Use in Elderly Men : Dosage adjustments are not required in elderly patients.

Children: Sildenafil is not indicated for use in children (<18 years).

Patients with Impaired Renal Function: Dosage adjustments are not required in patients with mild to moderate renal impairment (creatinine clearance = 30-80 mL/min).

Since sildenafil clearance is reduced in patients with severe renal impairment (creatinine clearance <30 mL/min), a 25-mg dose should be considered.

Patients with Impaired Hepatic Function: Since sildenafil clearance is reduced in patients with hepatic impairment (eg, cirrhosis), a 25-mg dose should be considered.

Patients Using Other Medicines: Given the extent of the interaction with patients receiving concomitant therapy with ritonavir, it is recommended not to exceed a maximum single dose of 25 mg sildenafil in a 48-hr period. A starting dose of 25 mg should be considered in patients receiving concomitant treatment with CYP3A4 inhibitors eg, erythromycin, saquinavir, ketoconazole, itraconazole (see Interactions).

In order to minimize the potential for developing postural hypotension, patients should be stable on α -blocker therapy prior to initiating sildenafil treatment. In addition, initiation of sildenafil at lower doses should be considered.

CONTRAINDICATIONS:

Known hypersensitivity to any component of Avogra.

Sildenafil was shown to potentiate the hypotensive effects of acute and chronic nitrates and its administration to patients who are concurrently using nitric oxide donors, organic nitrates or organic nitrites in any form either regularly or intermittently is therefore contraindicated.

Sildenafil is contraindicated in patients who have loss of vision in one eye because of non-arteritic anterior ischaemic optic neuropathy (NAION), regardless of whether this episode was in connection or not with previous PDE5 inhibitor exposure.

The safety of sildenafil has not been studied in the following sub-groups of patients and its use is therefore contraindicated: severe hepatic impairment, hypotension (blood pressure < 90/50 mmHg), recent history of stroke or myocardial infarction and known hereditary degenerative retinal disorders such as retinitis pigmentosa (a minority of these patients have genetic disorders of retinal phosphodiesterases).

PRECAUTIONS AND WARNINGS

A thorough medical history and physical examination should be undertaken to diagnose erectile dysfunction, determine potential underlying causes, and identify appropriate treatment.

There is a degree of cardiac risk associated with sexual activity; therefore, physicians may wish to consider the cardiovascular status of their patients prior to initiating any treatment for erectile dysfunction.

Agents for the treatment of erectile dysfunction should not be used in men for whom sexual activity is inadvisable.

Serious cardiovascular events, including myocardial infarction, sudden cardiac death, ventricular arrhythmia, cerebrovascular hemorrhage and transient ischemic attack have been reported post-marketing in temporal association with the use of sildenafil for erectile dysfunction. Most, but not all, of these patients had pre-existing cardiovascular risk factors. Many of these events were reported to occur during or shortly after sexual activity, and a few were reported to occur shortly after the use of sildenafil without sexual activity. Others were reported to have occurred hours to days after the use of sildenafil and sexual activity. It is not possible to determine whether these events are related directly to sildenafil, to sexual activity, to the patient's underlying cardiovascular disease, to a combination of these factors, or to other factors.

Sildenafil has been shown to have systemic vasodilator properties that result in transient decreases in blood pressure. This is of little or no consequence in most patients. However, prior to prescribing sildenafil, physicians should carefully consider whether their patients with certain underlying conditions could be adversely affected by such vasodilatory effects, especially in combination with sexual activity. Patients with increased susceptibility to vasodilators include those with left ventricular outflow obstruction (e.g., aortic stenosis, hypertrophic obstructive cardiomyopathy), or those with the rare syndrome of multiple system atrophy manifesting as severely impaired autonomic control of blood pressure.

Non-arteritic anterior ischemic optic neuropathy (NAION), a rare condition and a cause of decreased vision or loss of vision, has been reported rarely post-marketing with the use of all PDE5 inhibitors, including sildenafil. Most of these patients had risk factors such as low cup to disc ratio ('crowded disk'), age over 50, diabetes, hypertension, coronary artery disease, hyperlipidemia and smoking. An observational study evaluated whether recent use of PDE5 inhibitors, as a class, was associated with acute onset of NAION. In case of sudden visual loss, patients should be advised to stop taking sildenafil and consult a physician immediately.

Individuals who have already experienced NAION are at increased risk of NAION recurrence. Therefore physicians should discuss this risk with these patients and whether they could be adversely affected by use of PDE5 inhibitors. PDE5 inhibitors, including sildenafil, should be used with caution in these patients and only when the anticipated benefits outweigh the risks.

Caution is advised when sildenafil is administered to patients taking an alpha-blocker, as the co-administration may lead to symptomatic hypotension in a few susceptible individuals. In order to minimize the

potential for developing postural hypotension, patients should be hemodynamically stable on alpha-blocker therapy prior to initiating sildenafil treatment. Initiation of sildenafil at lower doses should be considered. In addition, physicians should advise patients what to do in the event of postural hypotensive symptoms.

A minority of patients with the inherited condition retinitis pigmentosa have genetic disorders of retinal phosphodiesterases. There is no safety information on the administration of sildenafil to patients with retinitis pigmentosa, therefore, sildenafil should be administered with caution to these patients.

Sildenafil potentiates the anti-aggregatory effect of sodium nitroprusside (a nitric oxide donor). There is no safety information on the administration of sildenafil to patients with bleeding disorders or active peptic ulceration, therefore sildenafil should be administered with caution to these patients.

Agents for the treatment of erectile dysfunction should be used with caution in patients with anatomical deformation of the penis (such as angulation, cavernosal fibrosis or Peyronie's disease), or in patients who have conditions which may predispose them to priapism (such as sickle cell anemia, multiple myeloma or leukemia).

Prolonged erections and priapism have been reported with sildenafil in post-marketing experience. In the event of an erection that persists longer than 4 hours, the patient should seek immediate medical assistance. If priapism is not treated immediately, penile tissue damage and permanent loss of potency could result.

The safety and efficacy of combinations of sildenafil with other PDE5 inhibitors, or other pulmonary arterial hypertension (PAH) treatments containing sildenafil, or other treatments for erectile dysfunction have not been studied, and the use of such combinations is not recommended.

No causal relationship has been made between the use of PDE5 inhibitors and sudden decrease or loss of hearing. In case of sudden decrease or loss of hearing patients should be advised to stop taking sildenafil and consult a physician promptly.

PREGNANCY:

Pregnancy and Lactation

Avogra 100mg Tablet is not indicated for use in newborns, children, or women.

DRUG INTERACTIONS:

Sildenafil metabolism is principally mediated by the cytochrome P-450 (CYP) isoforms 3A4 (major route) and 2C9 (minor route). Therefore, inhibitors of these isoenzymes may reduce sildenafil clearance.

There is a potential reduction in sildenafil clearance when co-administered with CYP3A4 inhibitors (eg, ketoconazole, erythromycin, cimetidine).

When a single 100-mg dose of sildenafil was administered with erythromycin, a specific CYP3A4 inhibitor, at steady state (500 mg twice daily for 5 days), there was a significant increase in sildenafil systemic exposure (AUC).

Co-administration of the HIV protease inhibitor saquinavir, a CYP3A4 inhibitor, at steady state (1200 mg thrice daily) with sildenafil (100-mg single dose) resulted in a significant increase in sildenafil C_{max} and an increase in sildenafil AUC. Sildenafil had no effect on saquinavir pharmacokinetics. Stronger CYP3A4 inhibitors eg, ketoconazole and itraconazole would be expected to have still greater effects.

Co-administration of the HIV protease inhibitor, ritonavir, which is a highly potent P-450 inhibitor at steady state (500 mg twice daily) with sildenafil (100 mg single dose) resulted in a 4-fold increase in sildenafil C_{max} and a 11-fold increase in sildenafil plasma AUC. At 24 hrs, the plasma levels of sildenafil were still approximately 200 ng/mL, compared to approximately

5 ng/mL when sildenafil was dosed alone. This is consistent with ritonavir's marked effects on a broad range of P-450 substrates. Sildenafil had no effect on ritonavir pharmacokinetics.

When the dose of sildenafil for subjects receiving potent CYP3A4 inhibitors was administered as recommended, there is potentiation in the free plasma sildenafil concentration

Single doses of antacid (magnesium hydroxide/aluminium hydroxide) did not affect the bioavailability of sildenafil.

No interaction was seen when sildenafil (100 mg) was co-administered with amlodipine in hypertensive patients. The mean additional reduction on supine systolic blood pressure was 8 mm Hg systolic and 7 mm Hg diastolic.

When the α -blocker doxazosin (4 mg) and sildenafil (25, 50 or 100 mg) were administered simultaneously to patients with benign prostatic hyperplasia (BPH), additional average reductions of supine blood pressure of 7 mmHg systolic and 7 mmHg diastolic were observed.

When higher doses of sildenafil and doxazosin (4 mg) were administered simultaneously, there were infrequent reports of patients who experienced symptomatic postural hypotension within 4 hours of dosing. Simultaneous administration of sildenafil to patients taking α -blocker therapy may lead to symptomatic hypotension in some patients.

SIDE EFFECT

The adverse events were generally transient and mild to moderate in nature.

In fixed-dose studies, the incidence of some adverse events increased with dose. The nature of the adverse events in flexible-dose studies, which more closely reflect the recommended dosage regimen, was similar to that for fixed-dose studies.

The very commonly adverse reaction is headache.

The common adverse reactions are:

- a) Nervous system disorders: dizziness
- b) Eye disorders: visual colour distortions and disturbance, blurred vision
- c) Respiratory, thoracic and mediastinal disorders: rhinitis (nasal congestion)
- d) Gastrointestinal disorders: nausea, dyspepsia
- e) Vascular disorders: vasodilation (flushing)

The uncommon adverse reactions are:

- a) Infections and infestations: rhinitis
- b) Immune system disorders: Hypersensitivity
- c) Nervous system disorders: somnolence, hypoaesthesia
- d) Eye disorders: lacrimation disorders, eye pain, photophobia, photopsia, ocular hyperemia, visual brightness, conjunctivitis.
- e) Ear and labyrinth disorders: vertigo, tinnitus
- f) Cardiac disorders: tachycardia, palpitation
- g) Respiratory, thoracic and mediastinal disorders: epistaxis, sinus congestion
- h) Vascular disorders: hypotension, hypertension
- i) Gastrointestinal disorders: gastro oesophageal reflux disease, vomiting, abdominal pain upper, dry mouth
- j) Skin disorders: rash
- k) Musculoskeletal disorders: myalgia, pain in extremity
- l) Renal and urinary disorders: haematuria
- m) General disorders: Chest pain, fatigue, feeling hot
- n) Heart rate increased

OVERDOSAGE:

In cases of overdose, standard supportive measures should be adopted as required. Renal dialysis is not expected to accelerate clearance as sildenafil is highly bound to plasma proteins and it is not eliminated in the urine.

STORAGE:

Store below 30°C. Protect from light and moisture.
Keep out of reach of children.

PACKING:

Pack of 4 (1 blister of 4 tablets), pack of 8 (2 blister of 4 tablets) and pack of 28 tablets (7 blisters of 4 tablets).

Manufactured Under License from

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